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Mutation in *pvcABCD* operon of *Pseudomonas aeruginosa* modulates MexEF-OprN efflux system and hence resistance to chloramphenicol and ciprofloxacin

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A B S T R A C T

Pseudomonas aeruginosa harbors *pvcABCD* operon that is responsible for the synthesis of paerucumarin. Here we report the involvement of *pvcABCD* operon in chloramphenicol and ciprofloxacin resistance. *P. aeruginosa* mutant defective in *pvcB* (PW4832) was more sensitive to chloramphenicol and ciprofloxacin in comparison with its parent strain (MPAO1). A mutation in *pvcA* gene in MPAO1 (PW4830) did not alter the sensitivity to either antibiotic. As chloramphenicol and ciprofloxacin are substrates of MexEF-OprN efflux pump, so we decided to investigate the modulation of MexEF-OprN and its transcriptional regulator MexT in PW4832, PW4830 and MPAO1 strains. We isolated and sequenced *mexT* gene from MPAO1, PW4830 and PW4832. The nucleotide sequence of *mexT* gene in all three strains was identical. Expression levels of *mexEF-oprN*, *mexT* and *mexS* genes were checked via quantitative real-time RT-PCR. All these genes showed significant repression in mRNA levels in PW4832 as compared to MPAO1. These results indicate that chloramphenicol and ciprofloxacin sensitivity in PW4832 is due to transcriptional repression of *mexT* and *mexEF-oprN* genes. Exogenous addition of paerucumarin resumed the expression of *mexT* and *mexEF-oprN* genes as well as resistance against chloramphenicol and ciprofloxacin in PW4832 strain. This is a novel finding linking *pvcB* gene of *P. aeruginosa* with chloramphenicol and ciprofloxacin resistance and MexEF-OprN pump modulation which needs to be further explored.

1. Introduction

Pseudomonas aeruginosa is an opportunist bacillus causing infections in immune compromised patients of acute leukemia, cancer, HIV and intravenous drug abusers [1]. This pathogen is equipped with an arsenal of virulence factors and a variety of tools for antibiotic resistance [2–4]. The wild type strains of *P. aeruginosa* (PAO1 and PA14) have been characterized to be sensitive to chloramphenicol (CAM) and ciprofloxacin (CP) but antibiotic resistant mutants of these strains have been isolated which overexpress MexEF-OprN efflux pump [5,6]. Chloramphenicol and ciprofloxacin are substrates for MexEF-OprN efflux pump and hence resistance to these antibiotics has been attributed to the activity of MexEF-OprN [6] and its regulators MexT and MexS [5]. PAO1 strain possess *mexT* gene which contains 8bp insertion, disrupting the open reading frame and encoding non-functional protein which cannot activate the efflux pump and renders bacteria sensitive to CAM and CP [7]. MPAO1 strain of *P. aeruginosa* has undergone a second mutation resulting in the deletion of 8bp insertion and functional MexT regulator. MPAO1 possesses functional MexEF-OprN efflux pump rendering it resistant to CAM and CP with minimum inhibitory concentration of 1024 μg/ml and 0.5 μg/ml respectively [8]. In this study, we have characterized a transposon mutant PW4832 of MPAO1 strain which was sensitive to CAM and CP.

PW4832 mutant possess a mutation in *pvcB* gene of *pvcABCD* operon [9] which is involved in the synthesis of paerucumarin [10]. In this operon, *pvcA* encodes an isonitrile synthase that produces isonitrile functionalized tyrosine (IFT) using tyrosine as a substrate [11]. *pvcB* gene produces an oxidoreductase [12,13] which oxidizes IFT to give an intermediate unsaturated compound. The *pvcC* and *pvcD* are monooxygenases which oxidize this intermediate compound to a catechol and dihydroxy phenols [10,14,15]. Intramolecular cyclization and carboxylation of this catechol generates isonitrile functionalized dihydroxy-cumarin or trivially known as paerucumarin [10]. Paerucumarin is an iron binding molecule [16,17] which plays role in biofilm formation by modulating fimbrial chaperon usher pathway [18]. Paerucumarin precursor (IFT) increased rhamnolipid production,

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swarming motility and quorum sensing in MPAO1 strain [19]. In this study, we tried to explore the link between chloramphenicol resistance and *pvcABCD* operon.

2. Materials and methods

2.1. Strains and culturing conditions

Three strains of *Pseudomonas aeruginosa* were used in the study and their details are enlisted in Table 1. Bacterial culturing was done using Luria-Bertani (LB) broth at 37 °C and 120 rpm for 16–18 h. Antibiotics were supplemented in the medium for each strain to avoid contamination. Parental strain MPAO1 was cultured by adding 200 μg/ml chloramphenicol and for *pvcA* and *pvcB* mutants, 60 μg/ml tetracycline was added to LB broth and agar plates. All culturing was done under sterile conditions in a BSL-1 laminar flow hood.

2.2. Antibiotic assay for chloramphenicol and ciprofloxacin

For the assessment of antibiotic resistance in all three test strains, sterile LB-broth was inoculated with glycerol stocks of *P. aeruginosa* strains and incubated at 37 °C and 120 rpm for 16–18 h. Cultures were sub cultured with an initial cell density of 0.02–0.04 in LB broth supplemented with required concentrations of chloramphenicol/ciprofloxacin. No antibiotic was added to one set of wells to compare the normal growth of all three test strains. The assay was setup in sterile, 24-well cell culture plates (SPL Life Sciences) in quadruplicates and plates were incubated for 16 h at 37 °C with shaking. Optical density (O.D. 600 nm) of cultures from all wells was measured via spectrophotometer (OPTIMA SP300 spectrophotometer). The values obtained were used to plot a graph and statistical analysis using Prism GraphPad.

2.3. Isolation and sequencing of mexT gene from P. aeruginosa

Overnight cultures of strains of *P. aeruginosa* were pelleted down by centrifugation and genomic DNA was extracted using phenol chloroform isolation method [21]. DNA was dissolved in 50 μl autoclaved distilled H_2O. DNA samples were treated with RNase at 37 °C for 30 min to remove RNA.

Using the forward and reverse primer set and genomic DNA as template, *mexT* gene was amplified via polymerase chain reaction (PCR) as described [18] from *P. aeruginosa* strains. For better quality sequence, PCR product was cloned in TA cloning vector pTZ57R/T (Thermo Fisher Scientific) and sequenced using M13 universal primers as previously described [22].

Table 1
Pseudomonas aeruginosa strains and plasmids used in the study.

P. aeruginosa Strains and plasmids	Description	Source of strain
Strains		
MPAO1	*P. aeruginosa* prototrophic laboratory strain, Cam^r	[9,20] UWGC
PW4830	*pvcA*-F05::IS*lacZ*/hah; out of frame fusion in MPAO1; Tet^r	[9] UWGC
PW4832	*pvcB*-G05::IS*lacZ*/hah; out of frame fusion in MPAO1; Tet^r	[9] UWGC
Plasmids		
pCR2.1–1.8	pCR®2.1-TOPO® carrying the 1.8-kbp *Pst*I *P. aeruginosa* stability fragment; Cb^r, Km^r	[18]
pLL2	pCR2.1–1.8 carrying intact *pvcB* on a 1200-bp fragment from MPAO1; Cb^r, Km^r	[18]

UWGC, University of Washington Genome Center; Cam, chloramphenicol; Tet, tetracyclin; Cb, carbenicillin; Km, kanamycin; ^{r}, resistant.

2.4. Gene expression by quantitative and semi-quantitative PCR

Bacterial cultures (16–18 h) of each strain were pelleted down in RNA protect Bacterial Reagent (Qiagen) according to manufacturer's instructions. RNA was extracted using RNeasy mini kit (Qiagen) with on-column DNase treatment as previously described [23]. Quantity and purity of extracted RNA was measured using spectrophotometer (Nanodrop) and quality was assessed through gel electrophoresis. Total RNA (5 μg) was reverse transcribed to cDNA using random hexamer primers and revertAid cDNA synthesis kit (Thermo Scientific). Gene specific primers (Table 2) were used for amplification in real-time RT-PCR using Sybr Green master mix (Thermo Scientific) and reaction was performed in PikoReal realtime thermocycler. Four replicates were used for each sample and data was analyzed using PikoReal Software 2.2. Melt curve analysis was performed to validate the specificity of amplicons and 30S rRNA(*rpsL*) was used as housekeeping gene for normalization of cDNA. Two-way ANOVA was used for statistical analysis and Bonferroni posttest was used for significance analysis.

3. Results

3.1. Functional inactivation of pvcB gene makes P. aeruginosa sensitive to chloramphenicol and ciprofloxacin

P. aeruginosa MPAO1 strain harbors *pvcABCD* operon which produces paerucumarin. Two pvc-operon mutants, PW4830 (*ΔpvcA*) and PW4832 (*ΔpvcB*) along with parent strain MPAO1 were tested for their resistance against chloramphenicol and ciprofloxacin. Chloramphenicol resistance of MPAO1, PW4830 and PW4832 was measured by adding various chloramphenicol concentrations (0, 6.25, 12.5, 25, 50, 100, 200 and 400 μg/ml) to bacterial cultures and cell density was measured after 8 h of incubation. We found that both MPAO1 and PW4830 strains had similar level of cell density while in PW4832, growth was significantly reduced (Fig. 1A). Results revealed that PW4832 is more sensitive to chloramphenicol as compared to the parent strain MPAO1 and PW4830. Growth of all three strains decreased with increasing antibiotic concentration until becoming almost zero at the concentration of 400 μg/

Table 2
Primers used in gene amplification and mRNA expression analysis.

Gene	Locus ID	Product Description	Primer ID	Sequence (5′-3′)
Full length gene amplification				
mexT	PA2492	Transcriptional regulator MexT	*mexT full*(F)	AAACCACCCGTCGTTATTGA
			mexT full(R)	ATGGACAGTGCTTTCGATGG
Gene expression				
rpsL	PA4268	30S ribosomal Protein	*rpsL* (F)	ACCAACGGTTTCGAGGTTTC
			rpsL (R)	CGACCCTGCTTACGGTCTT
mexT	PA2492	Transcriptional regulator MexT	*mexT* (F)	GTCGAGTTCGGCCTGTTG
			mexT (R)	TTGCGCTTGGCGTTG
mexE	PA2493	RND multidrug efflux membrane protein MexE	*mexE* (F)	TCCTCGACAAGGTCAA
			mexE (R)	ATCCAGGACCAGCACGAA
mexF	PA2494	RND multidrug efflux transporter MexF	*mexF* (F)	ATCAAGGTCAGCGACACCTC
			mexF (R)	TAGGTCAGCTCGGTCCACTC
oprN	PA2495	Multidrug efflux outer membrane protein OprN	*oprN* (F)	GGCCAGCTACGAACAGCA
			oprN (R)	TCCAGCAGCACCAGGAAA
mexS	PA2491	MexS	*mexS* (F)	CGTGAGCTGGAAGGATGTG
			mexS (R)	GGTATGGGCGGGGAAA

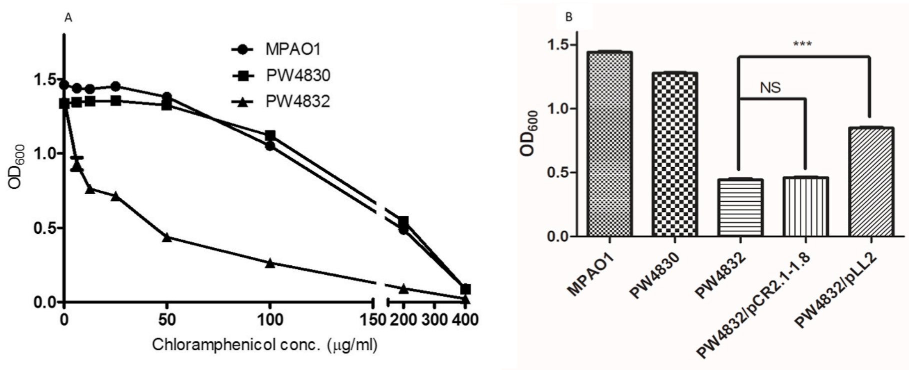

Fig. 1. PW4832 (pvcB mutant) is sensitive to chloramphenicol antibiotic as compared to parent strain MPAO1 of *P. aeruginosa.* Growth of PW4832 (ΔpvcB) strain of *P. aeruginosa* is inhibited as compared to MPAO1 and PW4830 (*ΔpvcA*) when grown in LB-medium supplemented with different concentrations of chloramphenicol (A) and the effect is partially complemented on addition of pLL2 plasmid containing *pvcB* gene (B) when grown in LB-medium supplemented with 25 μg/ml of chloramphenicol.

ml. Maximum difference between the cell densities was noted at antibiotic concentrations of 50 μg/ml and 100ug/ml as indicated in Fig. 1A. Transformation of pLL2 plasmid containing *pvcB* gene in PW4832 strain significantly improves the resistance to chloramphenicol as compared to pCR2.1–1.8 vector (Fig. 1B).

Resistance to ciprofloxacin is also significantly reduced in PW4832 as compared to MPAO1 and PW4830 (Fig. 2A). On complementation with pLL2 plasmid, the resistance to ciprofloxacin is enhanced 3.3 fold as compared to PW4832/pCR2.1–1.8 (Fig. 2B). Complementation of *pvcB* mutant with *pvcB* gene containing plasmid partially improves the resistance to chloramphenicol and ciprofloxacin but does not resume to the level of parent strain MPAO1.

3.2. Chloramphenicol sensitivity in PW4832 is not due to mutation of mexT gene

In *P. aeruginosa* transposon mutants, chloramphenicol sensitive strains have emerged from chloramphenicol resistant MPAO1 strain due to spontaneous mutations in *mexT* or *mexF* genes [24]. Other reports showed transformation of less virulent phenotype of MPAO1 strain to a more virulent phenotype by a spontaneous single nucleotide mutation in *mexT* gene [25]. PW4832 strain has been previously reported to produce higher levels of pyocyanin, rhamnolipids and quorum sensing signal [19]. In order to rule out the presence of unintended mutation in *mexT* or *mexF* genes of PW4832 or PW4830, we isolated the *mexT* and *mexF* genes from mutants and MPAO1 parent strains. Speculating the involvement of MexT in the chloramphenicol sensitivity of PW4832, genome sequence of *mexT*, a mutational hotspot in MPAO1 was analyzed in all three strains. Gene specific primers amplified bands of equal size (1165bp) in all three strains (Fig. 3A) nullifying the deletion or insertion of a large fragment in PW4832. The amplified bands were sequenced to find smaller or point mutations in strains under study.

DNA sequencing of *mexT* gene revealed no difference between MPAO1 and its mutants. The sequence homology was checked by comparing the nucleotide sequences of *mexT* gene from three strains with the PAO1 wild type and PA14 *mexT* sequences at www.pseudomonas.com. All the three test strains contained an 8-bp deletion at position 240 when aligned against *mexT* gene from *P. aeruginosa* PAO1 wild type and PA14 strains which are sensitive to chloramphenicol (Fig. 3B). No other mutation was detected in other parts of the gene sequence of the three strains under study. Thus, *mexT* gene in PW4832 and MPAO1 strain existed in its original functional form which is capable of activating *mexEF-oprN* operon and conferring chloramphenicol resistance. PA14 *mexT* gene was 121bp shorter than PAO1 and MPAO1 in 5' region pointing towards a possible difference in the regulation of *mexT* gene in both strains (www.pseudomonas.com) [26].

3.3. MexEF-OprN pump and its regulators are transcriptionally silenced in PW4832 strain of P. aeruginosa

Since no mutation was detected in the sequence of *mexT* gene, we analyzed the transcription level of *mexT* gene in MPAO1, PW4830 and PW4832 via quantitative real-time RT-PCR. The results indicated that *mexT* expression levels were comparable in MPAO1 and PW4830. However, Chloramphenicol sensitive mutant (PW4832) showed no amplification of *mexT* mRNA (Fig. 4A and B). The housekeeping gene 30S rRNA (*rpsL*) was amplified in all three strains at equal levels (Fig. 4A).

In addition to MexT regulator, the genes encoding MexEF-OprN pump were also checked for their relative expression among the three strains. mRNA accumulation of *mexE*, *mexF* and *oprN* transcripts was not detected in PW4832 relative to the parent MPAO1 and PW4830 strains (Fig. 4A and B). *mexS* gene which indirectly regulates *mexT* was also not detected in PW4832 mRNA library (Fig. 4A and B). However, no

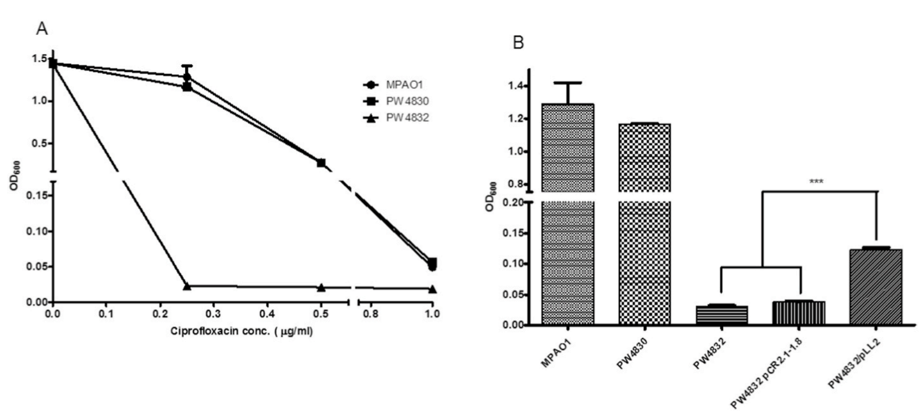

Fig. 2. PW4832 (*pvcB* mutant) is sensitive to ciprofloxacin antibiotic as compared to parent strain MPAO1 of *P. aeruginosa*. MPAO1 strain of *P. aeruginosa* loses its resistance against ciprofloxacin when *pvcB* gene is non-functional (PW4832) and effect is slightly complemented on addition of plasmid containing *pvcB* gene.

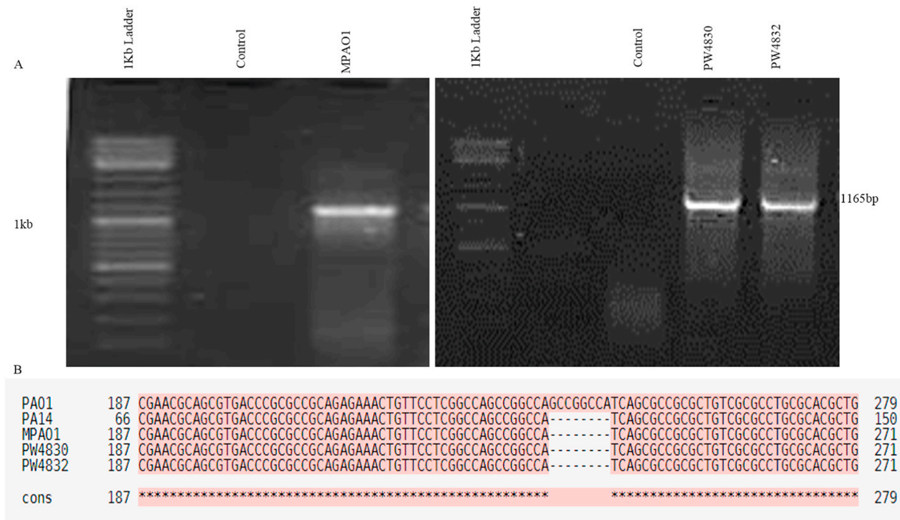

Fig. 3. *mexT* gene size and sequence is not altered in PW4832 as compared to MPAO1 parent strain of *P. aeruginosa*. PCR amplification of *mexT* gene in MPAO1, PW4830 and PW4832 showing same size (A). 8bp deletion is present in MPAO1, PW4830, PW4832 and PA14 but not in PAO1 (B). Lane labelled as control indicates no template control.

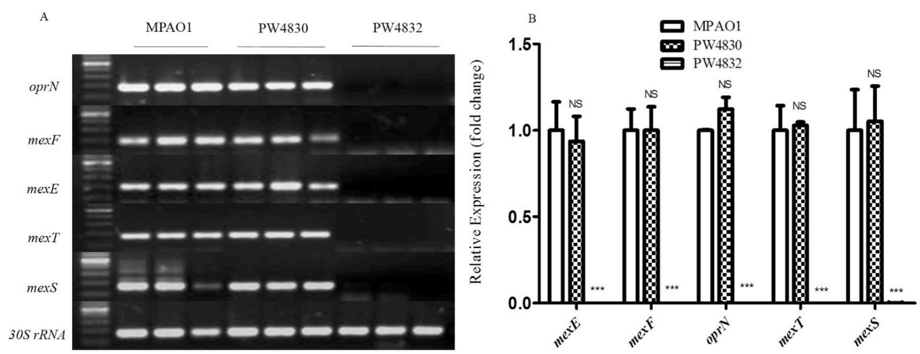

Fig. 4. *mexEF-oprN* system, *mexS* and *mexT* are transcriptionally repressed in PW4832. Semi-quantitative reverse transcriptase PCR(A) and relative expression using real-time RT-PCR (B) of *mexEF-oprN, mexS* and *mexT* genes showing transcriptional deactivation of MexEF-OprN efflux pump and its regulators (MexS and MexT) in PW4832 as compared to MPAO1 and PW4830. Error bars indicate standard deviation, NS indicates *p-value* > 0.05 and *** indicates p-value < 0.01.

significant difference was detected in the mRNA levels of PW4830 and MPAO1.These results confirmed that mutation in *pvcB* gene of *pvcABCD* operon transcriptionally modulates the expression of MexEF-OprN efflux pump and its regulatory genes *mexS* and *mexT*.

3.4. Exogenous addition of paerucumarin resumes the transcription of MexEF-OprN pump and its regulator but not IFT

In order to check the effect of IFT and paerucumarin (PC) on activation of MexEF-OprN pump, we isolated IFT and PC as previously described [18,19] and exogenously supplied to the bacterial cultures.

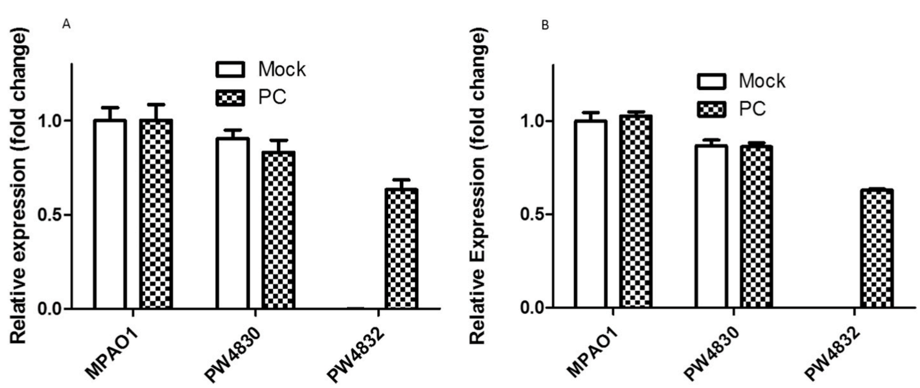

Fig. 5. Paerucumarin addition enhances the expression of MexEF-OprN efflux pump. Expression of *mexT* (A) and *mexE* (B) *genes* is significantly enhanced in PW4832 strain of *P. aeruginosa* grown in the LB medium containing exogenously added Paerucumarin (PC) as compared to solvent (Mock). Error bars represent standard deviation among replicates.

The expression of *mexT* (Fig. S1A) and *mexE* (Fig. S1B) genes was not significantly affected by the addition of IFT indicating that IFT does not affect MexEF-OprN pump directly. However, exogenous addition of PC to *P. aeruginosa* strains resumed the expression of *mexT* gene (Fig. 5A) and *mexE* gene (Fig. 5B) in PW4832 as compared to the cultures containing just the solvent of PC (Mock). The expression of *mexT* and *mexE* in MPAO1 and PW4830 was not affected significantly. These results indicate that PC resumes the expression of *mexEF-oprN* operon. In order to check the efflux function of MexEF-OprN, we checked the chloramphenicol and ciprofloxacin resistance in PW4832 strain in the presence and absence of PC. Our results indicated that chloramphenicol and ciprofloxacin resistance was increased 4.1-fold and 4.3-fold in PW4832 in the presence of PC (Fig. 6).

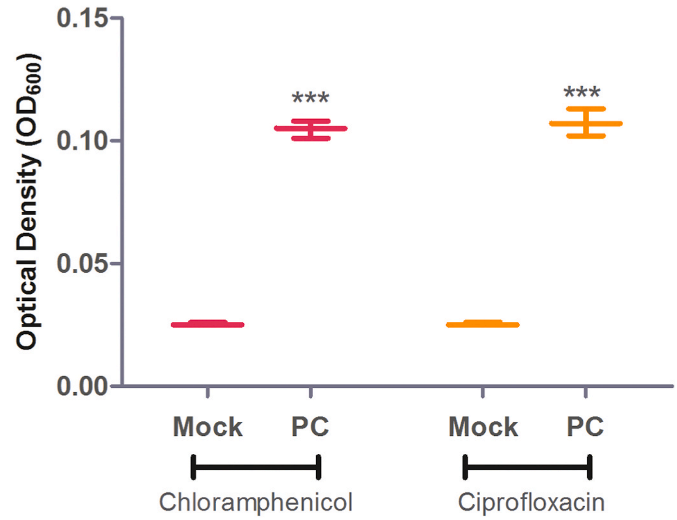

Fig. 6. Exogenous addition of paerucumarin improved chloramphenicol and ciprofloxacin resistance. Antibiotic resistance is significantly enhanced in PW4832 strains of *P. aeruginosa* grown in the LB medium containing exogenously added Paerucumarin (PC) as compared to solvent (Mock). Error bars represent standard deviation among replicates. *** indicates p-value < 0.01.

4. Discussion

MPAO1 strain of *P. aeruginosa* possesses an active MexT regulator and functional MexEF-OprN efflux pump which throws chloramphenicol and ciprofloxacin out of bacterial cells rendering them resistant to these antibiotics. Here we report that *pvcB* mutant of MPAO1 (PW4832) is sensitive to chloramphenicol and ciprofloxacin antibiotics in contrast with MPAO1 parent and complementation with *pvcB* gene improves the resistance to chloramphenicol and ciprofloxacin (Figs. 1 and 2) which is the manifestation of efflux by the MexEF-OprN pump. However, not to the level of MPAO1 parent indicating the involvement of other unknown factors. Among various multidrug efflux pumps operating in *P. aeruginosa* MPAO1 strain, the efflux of chloramphenicol from the bacterial cell is mainly controlled by MexEF-OprN efflux pump [27], its regulator MexT [28] located upstream of *mexEF-oprN* operon and *mexS* gene located upstream of *mexT* but in opposite direction [26,29]. Interestingly, *mexT*, *mexS* and *mexEF-oprN* operon are transcriptionally silent in *pvcB* mutant (Fig. 4) suggesting the involvement of a common regulator of MexT and MexS. Although this common regulator has not yet been identified but it was found that exogenous addition of PC resumes the expression of *mexT*, *mexEF-oprN* genes (Fig. 5) and resistance to chloramphenicol and ciprofloxacin (Fig. 6). We have compared the gene expression of all other RND efflux pumps through realtime RT-PCR and found that the expression levels of *mexAB-oprM*, *mexXY, mexCD-oprJ, mexGHI-opmD and mexJK* are comparable in all three strains under study (Data not shown).

MexEF-OprN pump has been reported to be involved in the transport of quorum sensing molecules like Pseudomonas quinolone signal (PQS) and its precursors [30]. We purpose that MexEF-OprN pump might be involved in maintaining a specific level of PC in the cell through an unknown regulator. Relative quantities of PC and its precursor IFT might regulate MexEF-OprN efflux pump through a common regulator of *mexT* and *mexS* (as both these regulators are transcriptionally quiescent in PW4832). In PC mutant (PW4832) strain which produces the precursor of paerucumarin (IFT) in excessive amounts, *mexT* is transcriptionally inactivated by the interaction of regulator and IFT. Thus silencing *mexEF-oprN* operon and closing the pump to prevent the export of paerucumarin precursor. In the parent MPAO1 which produces both IFT and PC and PW4830 which doesn't produce IFT or PC, the precursor and PC are in equilibrium rendering MexEF-OprN pump functional. When PW4832 cultures were supplemented with purified PC, *mexT* and *mexEF-oprN* were transcriptionally activated (Fig. 5) and efflux of chloramphenicol and ciprofloxacin was evident (Fig. 6). Addition of PC to IFT rich strain (PW4832) might have established the equilibrium between precursor and PC which is required for the activation of MexEF-OprN pump. Addition of IFT to MPAO1 and PW4830 increased IFT in comparison with PC and slightly reduced the expression of *mexE* and *mexT* genes (Fig S1) through modulating the regulator. These results clearly prove the involvement of *pvc* operon and PC in the functioning of MexT, MexS, MexEF-OprN efflux pump and antibiotic resistance. However, further investigations are needed to elucidate the molecular mechanism involved in the interplay of IFT, PC, common regulator and efflux pumps.

MexT transcriptional regulator has been reported to negatively regulate the production of virulence factors (PQS, pyocyanin and rhamnolipids) [25,30]. We previously reported the enhanced production of virulence factors in PW4832 strain as compared to MPAO1 [19]. Production of virulence factors is also comparable in PW4830 and parent strain MPAO1. In a previous study, we have found that PC mutants were deficient in biofilm and fimbriae production [18]. Extracellular addition of PC enhanced the expression of fimbrial chaperon usher pathway (cup) genes which enhance biofilm formation [18]. Biofilm cells are much more resistant to antibiotics as compared to planktonic cells [31,32]. PW4832 strain which is susceptible to chloramphenicol and ciprofloxacin was also found to be deficient in biofilm production. Thus PC enhances biofilm production and confers resistance to antibiotics. More research is needed to thoroughly investigate this link.

Author statement

Mutation in *pvcABCD* operon of *Pseudomonas aeruginosa* modulates chloramphenicol resistance and mexEF-OprN efflux system, AI and AA performed experiments. AM, MA and GS designed the experiments. NR provided technical advice and facilities for research. UQ designed the experiments, analyzed data and wrote paper.

Author's contribution

Anam Iftikhar and Azka Asif performed experiments. Asma Manzoor, Ghulam Sarwar and Muhammad Azeem designed the experiments. Naeem Rashid provided technical advice and facilities for research. Uzma Qaisar designed the experiments, analyzed data and wrote paper.

Declaration of competing interest

We declare that "There is no conflict of interest".

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.micpath.2020.104491.

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